

Overview

`sunlineEphem` computes the Sun direction unit vector relative to the spacecraft in the body frame based on ephemeris data. This direction vector can be used as a reference for downstream guidance and control modules.

The module layer (`SunlineEphem`) implements the `SysModel` interface, wiring messages and managing module lifecycle. The algorithm (`SunlineEphemAlgorithm`) contains the stateless computation that can be used stand-alone when the caller provides position and attitude data directly.

Module Inputs and Outputs

Module inputs:

Table 1: Module Input Messages

Msg Variable Name	Msg Type	Description
<code>sunPositionInMsg</code>	<code>EphemerisMsgF32Payload</code>	Read every <code>updateState()</code> . Contains the Sun's position expressed in the inertial frame.
<code>scPositionInMsg</code>	<code>NavTransMsgF32Payload</code>	Read every <code>updateState()</code> . Contains the spacecraft's position expressed in the inertial frame.
<code>scAttitudeInMsg</code>	<code>NavAttMsgF32Payload</code>	Read every <code>updateState()</code> . Contains the spacecraft's attitude (MRPs), used to rotate the Sun vector from the inertial frame to body frame.

Module output:

Table 2: Module Output Messages

Msg Variable Name	Msg Type	Description
<code>navStateOutMsg</code>	<code>NavAttMsgF32Payload</code>	Ephemeris-based Sun direction unit vector expressed in the body frame, written to <code>vehSunPntBdy</code> .

Algorithm Inputs and Outputs

Algorithm inputs:

Variable Name	Description
r_SN_N	Sun position vector expressed in the inertial frame ${}^N\mathbf{r}_{S/N}$.
r_BN_N	Spacecraft position vector expressed in the inertial frame ${}^N\mathbf{r}_{B/N}$.
sigma_BN	Spacecraft attitude MRPs $\boldsymbol{\sigma}_{BN}$, used to rotate the Sun direction from the inertial frame to the body frame.

Algorithm output:

Variable Name	Description
rHat_SB_B	Sun direction unit vector expressed in the body frame ${}^B\hat{\mathbf{r}}_{S/B}$. Returns the zero vector if the spacecraft and Sun are colocated.

Algorithm Assumptions

- All input position vectors must be expressed in the same inertial reference frame.

Algorithm Description

Given Sun position ${}^N\mathbf{r}_{S/N}$, spacecraft position ${}^N\mathbf{r}_{B/N}$, and spacecraft attitude MRPs $\boldsymbol{\sigma}_{BN}$:

1. Compute the Sun position relative to the spacecraft in the inertial frame:

$${}^N\mathbf{r}_{S/B} = {}^N\mathbf{r}_{S/N} - {}^N\mathbf{r}_{B/N}$$

2. Normalize the relative vector, build the DCM from the spacecraft attitude MRPs, and rotate the unit direction into the body frame:

$${}^B\hat{\mathbf{r}}_{S/B} = \text{normalize}([BN] {}^N\hat{\mathbf{r}}_{S/B})$$

Normalization uses a numerically stable algorithm, `stableNormalized()`, that returns the zero vector if ${}^N\mathbf{r}_{S/B}$ is zero.

The algorithm returns a unit direction vector for distinct positions. For colocated positions (truly identical positions), the zero vector is returned as a sentinel. Downstream consumers should check for this case.

Figure 1: SunlineEphem computes the inertial-frame Sun direction vector ${}^N\mathbf{r}_{S/B}$.

Module vs. Algorithm Usage

- Module (`SunlineEphem`, `SysModel`): use when operating inside the messaging framework. Connect `sunPositionInMsg`, `scPositionInMsg`, and `scAttitudeInMsg`, call `InitializeSimulation()/ExecuteSimulation()`, and read `navStateOutMsg`.
- Algorithm (`SunlineEphemAlgorithm`): use when you have position and attitude data directly. Call the static `update(r_SN_N, r_BN_N, sigma_BN)` method, which returns the body-frame Sun direction vector.